S10. Certified periodic orbits of a specified three-body model

Pure Thought research portfolio

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Source: synthesis challenge 27; original synthesis. This topic is distinct from, or adds a distinct objective to, the canonical PRD catalog. Crosswalk

Plain-language guide

The problem in everyday terms

A simulated three-body trajectory may nearly return to where it started, but a small numerical error can disguise whether an exactly repeating orbit exists. This problem converts a promising numerical trajectory into a proof that a nearby periodic solution exists and stays away from collisions, then examines how small disturbances behave over one cycle.

Key terms

- **Three-body problem** — The motion of three mutually gravitating bodies in the specified Newtonian model.
- **Restricted model** — A version in which one body has negligible mass and does not affect the other two.
- **Periodic orbit** — A trajectory that returns to the same state after a fixed positive time.
- **Shooting method** — Adjusting an initial state until its evolved state satisfies the return conditions.
- **Phase condition** — A rule fixing the arbitrary starting point along a periodic trajectory.
- **Validated integration** — Numerical evolution with guaranteed enclosures for errors.
- **Collision exclusion** — A proved positive lower bound on relevant body separations along the orbit.
- **Floquet multiplier** — An eigenvalue describing linear perturbation growth or rotation after one orbit.
- **Neutral direction** — A direction with constrained or trivial linear behavior arising from symmetries or conservation laws.

Why this matters

Periodic orbits are organizing reference motions in gravitational dynamics. Certifying their existence gives subsequent studies of nearby motion a dependable starting point and separates exact solutions from visual patterns in a simulation.

What progress would mean

Progress would enclose an orbit and its period, exclude collisions, and classify the reduced linear stability when the bounds permit. Linear stability would not alone prove nonlinear or indefinite stability, and a periodic-orbit proof would not establish connecting trajectories between different orbits.

Revised core question

Can a numerically proposed periodic orbit be enclosed by a validated shooting proof and classified by its reduced Floquet spectrum?

Precise scope, target and outputs

Choose one nondimensional Newtonian three-body model with positive rational masses, or a planar circular restricted model with fixed mass ratio. Declare collision exclusions and conserved quantities. Define a Poincaré section or phase condition to remove time translation, and reduce continuous symmetries. Solve the resulting shooting residual using a validated flow integrator and interval Newton/Krawczyk bounds for initial conditions and period. Integrate variational equations with enclosures for the monodromy matrix; remove the expected neutral directions before interpreting multipliers.

Detailed critique of the synthesis framing

The original bundles orbit existence, complete family discovery, linear stability, invariant manifolds and heteroclinic connections into one certificate. A small shooting residual is not existence, and stable-looking Floquet multipliers are not nonlinear stability. Near-return trajectories do not establish a connection between invariant manifolds. The revised first target is a single orbit with a phase-fixed existence proof; continuation, manifold enclosure and connection proofs are explicit later stages. This also preserves the figure-eight idea incorrectly folded into PRD 27's central configurations.

Validation and milestones

Baseline: certify one supplied orbit and its period. Strong: validated continuation over a parameter interval and a reduced Floquet classification. A manifold connection requires additional covering/intersection proof, not a plot of approaching trajectories.

Use the shared evidence standard. Unsuccessful search is unresolved unless a complete exclusion proof is supplied. These are proposed baseline specifications, not completed scientific results.

Displayed formulation and claim boundary

Supplemental specification.

$$F(z_0, T) = \varphi_T(z_0) - z_0 = 0, \qquad M = D\varphi_T(z_0).$$

The shooting residual must first be augmented and reduced using a phase condition, conserved quantities and continuous symmetries so the validated root problem has the correct dimension. Interval flow and root bounds establish existence; the reduced Floquet multipliers classify linear behavior. Neither a near return nor stable-looking multipliers prove nonlinear stability or a manifold connection.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in

this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.